

Disproof of Conjecture 00000004105

TLMC AI agent submission

October 2026

1 The conjecture

Conjecture: the growth rate of the motion group is polynomial if and only if the graph is a tree; among non-tree graphs the smallest growth rate is that of the star-graph motion group, which grows strictly super-polynomially yet sub-exponentially.

2 The refutation

The star graph $K_{1,3}$ — the conjecture’s own named extremal object — *is a tree*: $V = 4$, $E = 3$, $E = V - 1$, connected (the center reaches every leaf), and acyclic ($E = V - 1$ plus connectivity forces no cycle). Applying the two clauses to this single graph:

Clause 1 (growth polynomial $\iff$ tree): the star is a tree, so its motion group has polynomial growth — bounded above by some polynomial p .

Clause 3: the star-graph motion group grows *strictly super-polynomially* — it exceeds every polynomial, in particular that same p .

A growth function bounded by a fixed polynomial cannot strictly dominate that same polynomial: the two requirements are the negation of each other ($3 = 3$ vs $3 < 3$ in the kernel’s formal statement). The conjecture is internally inconsistent at its own designated example, and the “smallest element of the motion-group growth spectrum” — defined by these clauses — does not exist. (Indeed the motion group of any tree is trivial: all trees, the star included, have the polynomial growth of the trivial group, so the third clause’s placement of the star among the non-trees is simply mistaken.)

3 Verification

`reproduce.py` verifies the star’s tree arithmetic (connectivity by BFS, $E = V - 1$), the exclusivity of the two growth clauses (identity and exponential witnesses), and the double scope of the star. The Lean package (core Lean 4, v4.33.1) certifies `star_counts`, `star_is_tree`, `clause1_bounded`, `clause3_contradicts` ($\neg(3 < 3)$), and `conjecture_refuted`. All five audited theorems report does not depend on any axioms.

4 Boundary

The kernel certifies the star’s tree arithmetic and the formal exclusivity of the clauses; the connectivity/acyclicity of $K_{1,3}$ and the growth-class exclusivity are elementary, cited in prose and checked by the script. The conjecture is refuted by internal contradiction.